Hugh Morison

Photonic ICs and electronic-photonic co-design

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Education

Queen's University, PhD in Applied Physics – Kingston, ON — Sept 2019 – present

Thesis (in preparation): design, simulation, and experimental demonstration of analog optical computing platforms.

- **Silicon photonics:** contributed to multiple tapeouts on the AMF silicon photonic platform, from chip-level architecture and layout through packaging and post-fabrication test.
- **Simulation:** built system-level electronic-photonic co-simulation models (Verilog-A, Cadence, Python), parametrised by component-level simulation of the microring modulators and analog weight banks they contain.
- **Experiment:** built a high-baud-rate thin-film lithium niobate testbed from discrete components with collaborators, demonstrating a programmable 200 GOPS photonic Ising machine, published in *Nature*.
- **Hardware:** mixed-signal and RF PCB design, chip packaging and wirebonding, and automated optical/RF test and measurement.

Queen's University, BASc in Engineering Physics (Computing Option) – Kingston, ON — Sept 2015 – Apr 2019

Experience

Research Engineer, Hartley Ultrafast, Ltd. – Bristol, UK — Mar 2026 – present

- Own the chip-top design of an ongoing tapeout on the GlobalFoundries 45SPCLO monolithic CMOS-photonics platform, integrating the analog, digital, and photonic blocks of the wider design team into one verified top level in Synopsys OptoCompiler.
- Designed the microring resonance-locking circuit: a transimpedance amplifier on the modulator drop port drives a comparator against a DAC-set reference, giving a 1-bit converter that an integral controller uses to hold each ring at its setpoint; verified in PrimeSim.
- Lead system-level floorplanning and routing across the electronic and photonic layers, reconciling waveguide routing constraints with signal and power distribution.
- Implement the digital control logic through the custom analog design flow, covering for the absence of a dedicated digital designer.
- Built the demonstrator's calibration stack: a lithium niobate modulator bias controller, a fix for temperature drift in the optical bias points, and Bayesian system identification of every physical parameter with uncertainties.

Senior Optical Engineer, Milkshake Technologies, Inc. – Kingston, ON — Aug 2024 – Dec 2024

- Ran technical due diligence on commercialising the *Nature* photonic Ising machine, and showed that solution success rate decreases exponentially with problem size.
- Modelled compute efficiency per unit area for the thin-film lithium niobate architecture and found no competitive route, which informed the decision to stop the program.
- Extended that scaling analysis into ongoing work on the limits of hypermultiplexed photonic tensor cores.

Projects

fairchild — May 2026 – present

Open-source electronic-photonic circuit simulator, in development

- Modified nodal analysis engine in Rust that solves optical fields and electrical unknowns in the same Newton iteration, removing the fixed-point loop between separate photonic and electronic tools.
- Runs Verilog-A device models compiled to OSDI with transient noise, so a full link simulates from laser to transimpedance amplifier in one deck.

Skills

Photonic Integrated Circuits: Foundry tapeout on AMF silicon photonics, GlobalFoundries 45SPCLO monolithic CMOS-photonics, and thin-film lithium niobate; microring modulators and weight banks; layout, floorplanning, waveguide routing, DRC/LVS

Electronics: Analog mixed-signal design (TIA, comparator, DAC, closed-loop control), digital logic in a custom design flow, RF & mixed-signal PCB design, FPGA/RFSoC data acquisition

Modelling & Simulation: System-level electronic-photonic co-simulation, Verilog-A device models, FDTD & photonic circuit simulation, Bayesian inference & system identification, nonlinear dynamics

Test & Measurement: Fibre-coupled PIC characterisation, optical spectrum analysers, high-speed oscilloscopes & arbitrary waveform generators, automated measurement in Python, wirebonding, packaging, UV lithography

EDA: Synopsys OptoCompiler & PrimeSim, Cadence Virtuoso, KLayout, Ansys Lumerical, LTspice, KiCAD, Vivado

Programming & Tools: Python (NumPy/SciPy), C, C++, MATLAB, Verilog & Verilog-A, Git, Linux, LaTeX

Awards

Mitacs Accelerate Fellowship (Huawei Canada): Industry-partnered research fellowship, $45,000, 2022

Ontario Graduate Scholarship: Provincial merit award, $15,000, 2024

Outreach & Teaching

Scaling and Efficiency Metrics of TFLN Photonic Ising Machines — Feb 2026
Guest lecture, ETH Zurich (remote)

Thinking in Hardware: The history and future of neuromorphic computing — Oct 2025
Public talk, Queen's Society of Optics

Teaching Assistant, Queen's University — Sept 2019 – Dec 2025
Analog and digital electronics, engineering physics design projects, engineering physics laboratories

Selected Publications

Programmable 200 GOPS Hopfield-inspired photonic Ising machine — Dec 2025
Al-Kayed, St-Arnault, ***Morison***, et al.

Co-first author, equal contribution.
www.nature.com/articles/s41586-025-09838-7 (Nature 648, 576–584)

Photonic fully-connected hybrid beamforming using microring weight banks — Nov 2025
Nichols, ***Morison***, et al.
www.nature.com/articles/s44172-025-00532-0 (Communications Engineering 4, 201)

120 GOPS Photonic Tensor Core in Thin-Film Lithium Niobate for Inference and In Situ Training — Oct 2024
Lin, et al.
www.nature.com/articles/s41467-024-53261-x (Nature Communications 15, 9081)

Nonlinear dynamics in neuromorphic photonic networks: Physical simulation in Verilog-A — Mar 2024
Morison, et al.
journals.aps.org/prapplied/abstract/10.1103/PhysRevApplied.21.034013 (Physical Review Applied)

Scaling of hypermultiplexed photonic tensor cores — 2026
Morison, et al.
(Manuscript in preparation)